THE STACK OF RELATIVELY PERFECT COMPLEXES IS FIBRED IN GROUPOIDS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Situation situation-complexes (canonical tag 0DLE). cited at statement line 5389. Standing situation.
- Definition categories-definition-fibred-groupoids (canonical tag 003T). cited at proof line 5397. The two conditions.

Source reference. Original definition 0DLE: sources/stacks-chow/tags.txt, source lines 17044–17044. Cited at statement line 5389: standing situation.

Source reference. Original definition 003T: sources/stacks-chow/tags.txt, source lines 147–147. Cited at proof line 5397: the two conditions.

Lemma 0.2. *In Situation 0DLE the functor $p : \mathit{Complexes}_{X/B} \longrightarrow (Sch/S)_{fppf}$ is fibred in groupoids.*

Proof. We show that p is fibred in groupoids by checking conditions (1) and (2) of Categories, Definition 003T. Given an object (T', g', E') of $\mathit{Complexes}_{X/B}$ and a morphism $h : T \to T'$ of schemes over S we can set $g = h \circ g'$ and $E = L(h')^*E'$ where $h' : X_T \to X_{T'}$ is the base change of h. Then it is clear that we obtain a morphism $(T, g, E) \to (T', g', E')$ of $\mathit{Complexes}_{X/B}$ lying over h. This proves (1). For (2) suppose we are given morphisms

$$(h_1, \varphi_1) : (T_1, g_1, E_1) \to (T, g, E) \quad \text{and} \quad (h_2, \varphi_2) : (T_2, g_2, E_2) \to (T, g, E)$$

of $\mathit{Complexes}_{X/B}$ and a morphism $h : T_1 \to T_2$ such that $h_2 \circ h = h_1$. Then we can let φ be the composition

$$L(h')^*E_2 \xrightarrow{L(h')^*\varphi_2^{-1}} L(h')^*L(h_2)^*E = L(h_1)^*E \xrightarrow{\varphi_1} E_1$$

to obtain the morphism $(h, \varphi) : (T_1, g_1, E_1) \to (T_2, g_2, E_2)$ that witnesses the truth of condition (2). □

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